

# Ziseok Lee



Personal Website: <https://ziseoklee.github.io/>  
Google Scholar: [https://scholar.google.com/citations?user=C3x\\_tewAAAAJ&hl=en](https://scholar.google.com/citations?user=C3x_tewAAAAJ&hl=en)  
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Integrated Masters with PhD Program  
Department of Biomedical Sciences  
Seoul National University

## Research Interests

Generative Models, AI4Science, Computer Vision

## Education

|                                                                                                |                   |
|------------------------------------------------------------------------------------------------|-------------------|
| Seoul National University                                                                      | 2025.03 - present |
| Masters-PhD Program in Biomedical Sciences advised by Prof. Kyungsu Kim                        |                   |
| Seoul National University                                                                      | 2021.03 - 2025.02 |
| B.S. in Mathematical Sciences (Major), B.S. in Computer Science and Engineering (Double Major) |                   |
| GPA 4.22/4.30 (Summa Cum Laude)                                                                |                   |

## Research Experience

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|-------------------------------------------------------------------------------------------------------|-------------------|
| ▼ Student Researcher at <b>AIBL</b>                                                                   | 2024.10 - present |
| Probabilistic Generative Models for Biomolecule Generation and Drug Design                            |                   |
| ▼ Undergraduate Research Assistant at <b>SNU MLLAB</b>                                                | 2024.07 - 2024.09 |
| Blackbox Optimization                                                                                 |                   |
| ▼ Undergraduate Research Assistant at <b>CTA Lab</b>                                                  | 2024.01 - 2024.08 |
| NP-hard Subgraph Matching Algorithms with Safety Conditions for Reduced Candidate Space               |                   |
| ▼ Undergraduate Research Assistant at <b>HYKE Group</b>                                               | 2022.12 - 2023.07 |
| Collective Motion in the Vicsek Particle Model, Hyperbolic Conservation Laws with Nonlocal Relaxation |                   |

## Awards & Scholarships

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|--------------------------------------------------------------------------|-----------|
| AI Seoul Tech Graduate Scholarship (Seoul Scholarship Foundation)        | 2025      |
| B.S. Degree Honors: Summa Cum Laude (Seoul National University)          | 2025      |
| Presidential Science Scholarship (Korea Student Aid Foundation)          | 2023-2024 |
| Dean's List (Department of Mathematical Sciences, SNU)                   | 2023-2024 |
| Professor Heo Sik Scholarship (Department of Mathematical Sciences, SNU) | 2022      |
| Gwanak Foundation Scholarship (Gwanak Foundation)                        | 2021      |

## Teaching Experience

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|--------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ▼ Teaching Assistant for <b>[Data Mining and Machine Learning]</b>                                                             | 2026.03 - 2026.06 |
| Assisted Professor Kyungsu Kim in teaching a third-year undergraduate course, supporting students through discussion sessions. |                   |
| ▼ Teaching Assistant for <b>[Introduction to Data Science]</b>                                                                 | 2025.03 - 2025.06 |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Assisted Professor Kyungsu Kim in teaching a second-year undergraduate course, supporting students through discussion sessions.                                                                                  |                   |
| ▼ Teaching Assistant for <b>[Calculus 1]</b>                                                                                                                                                                     | 2024.03 - 2024.06 |
| Taught a course on Calculus 1. Was awarded <b>Excellent TA</b> and gave a case presentation.                                                                                                                     |                   |
| ▼ Student-Directed Seminar <b>[Understanding the Brain as a Complex System]</b>                                                                                                                                  | 2023.09 - 2023.12 |
| Organized and taught a student-directed seminar integrating theoretical neuroscience, network science, deep learning, biology, and psychology to understand the brain as a complex, dynamical, entangled system. |                   |
| ▼ Peer Tutoring <b>[Calculus Tutoring for First Year Students]</b>                                                                                                                                               | 2023.03 - 2023.06 |
| Taught a tutoring class for "Differential and Integral Calculus 1" during the Spring semester of 2023.                                                                                                           |                   |

Grants & Funding

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| ▼ Strategic Hub for International Research Collaboration (SNU Office of Research Affairs)                                                                                                                                                                                                                                    | 2026 - 2028 |
| <b>Project Title:</b> An International Research Hub for Developing a 4D Molecular Generative AI Platform to Control Dynamic Protein Interaction Networks.                                                                                                                                                                    |             |
| <b>Funding:</b> ~\$380k (3 Years)                                                                                                                                                                                                                                                                                            |             |
| <b>Description:</b> Authored the grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim). I lead the design of stochastic interpolant-based diffusion models for drug optimization and protein conformational ensemble prediction. |             |

Publications

\* Equal contribution, † Corresponding Author

Conference Papers

- [First Author]** *On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering*, Ziseok Lee\*†, Minyeong Hwang\*, Wooyeol Lee, Sanghyun Jo, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang†, Kyungsu Kim†, Forty-third International Conference on Machine Learning (ICML) (2026)
- [Co-first Author]** *TRACE: Your Diffusion Model is Secretly an Instance Edge Detector*, Sanghyun Jo\*, Ziseok Lee\*, Wooyeol Lee, Jonghyun Choi, Jaesik Park†, Kyungsu Kim†, The Fourteenth International Conference on Learning Representations (**ICLR Oral, top 1.1%**) (2026)
- [Co-first Author]** *Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo Guidance in Discrete Diffusion*, Jaehyeon Kim\*, Ziseok Lee\*, Siyoung Choi, Kyungsu Kim†, (2026)
- [Co-first Author]** *Early Timestep Zero-Shot Candidate Selection for Instruction-Guided Image Editing*, Joowon Kim\*, Ziseok Lee\*, Donghyeon Cho, Sanghyun Jo, Yeonsung Jung, Kyungsu Kim†, Eunho Yang†, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2025)

Workshop Papers

- [Coauthor]** *HybridLinker: Topology-Guided Posterior Sampling for Enhanced Diversity and Validity in 3D Molecular Linker Generation*, Minyeong Hwang, Ziseok Lee, Kwang-Soo Kim, Kyungsu Kim†, Eunho Yang†, ICML 2025 Generative AI and Biology (GenBio) Workshop (2025)

Preprints

- [Co-first Author]** *ISAC: Training-Free Instance-to-Semantic Attention Control for Improving Multi-Instance Generation*, Sanghyun Jo\*, Wooyeol Lee\*, Ziseok Lee\*, Kyungsu Kim†, arXiv preprint arXiv:2505.20935 (2025)

Academic Services

[2025.04] Assisted Professor Kyungsu Kim in reviewing submissions in the field of computer vision for ICCV 2025.

[2025.03] Assisted Professor Kyungsu Kim in reviewing two submissions in the field of flow matching for ICML 2025.

**Biography**

Ziseok Lee is a graduate student in the [Department of Biomedical Sciences](#) at [Seoul National University](#). He is a member of the [Artificial Intelligence and Biomedical Informatics Lab \(AIBL\)](#) under the supervision of Professor Kyungsu Kim. He received his B.S. degree in the [Department of Mathematical Sciences](#) with a double major in the [Department of Computer Science and Engineering](#) at Seoul National University in 2025. His research interests include generative models (e.g., stochastic interpolants) and their applications to computer vision and biomedical science, particularly molecule generation for drug design.

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**Technical Skills**

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, React

Tools & Platforms: Git, Docker, Linux

Languages: Korean (native), English (fluent)

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**References**

**Dr. Kyungsu Kim** (Primary advisor)

Professor, School of Transdisciplinary Innovations

Seoul National University

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